

Feature Extraction Method of Motor Imagery EEG Based on DTCWT Sample Entropy

MENG Ming, LU Shaona, MAN Haitao, MA Yuliang, GAO Yunyuan

Institute of Intelligent Control and Robot, Hangzhou Dianzi University, Hangzhou 310018, China
E-mail: mnming@hdu.edu.cn

Abstract: Aiming at the applications of Brain Computer Interface (BCI) based on motor imagery EEG, this paper presents a feature extraction method combining Dual-Tree Complex Wavelet Transform (DTCWT) and sample entropy. Firstly, the motor imagery EEG signals are decomposed by DTCWT. Then, the rhythm waves corresponding to the event-related desynchronization (ERD)/event-related synchronization (ERS) phenomenon are extracted and reconstructed. Finally, the features are extracted from the rhythm signal using sample entropy. The Datasets1 of BCI Competition IV, including motor imagery EEG data of left hand, right hand and foot, is used to verify the proposed method. A Support Vector Machine (SVM) classifier is introduced in the classification experiments. The average classification accuracy rate of four subjects is 87.25% in the experiments with the Datasets1 of BCI Competition IV. The results show that the feature extraction method has more obvious separability and practicability.

Key Words: EEG; DTCWT; Sample Entropy; SVM

1 Introduction

Brain computer interface (BCI) is a communication system which does not depend on the brain's normal output pathways (i.e., peripheral nervous system and muscle tissue), to establish direct communication and control channel between brain and computer or other external equipment. The cerebral electric activity can be recorded via electroencephalography (EEG). The acquired EEG signals are converted into a control signal through feature extraction and recognition. People can directly express their own ideas or operate the external auxiliary motion equipment through the brain, with the wide application prospect in the field of brain science, rehabilitation engineering, biomedical engineering, entertainment and so on^[1-2]. According to the type of the applied EEG signals, BCIs fall into two classes: evoked EEG and spontaneous EEG. Motor imagery EEG is one type of the spontaneous EEG, which does not need external stimulation, and can realize the asynchronous communication.

For the imagined task based BCI, the key technology is how to carry out effective feature extraction and correct classification. Currently, the feature extraction of motor imagery EEG is mainly according to the ERD and ERS phenomenon of sensorimotor cortex signal while imagining, and this phenomenon is mainly concentrated in the alpha rhythm (8~15Hz) and the beta rhythm (15~30Hz) band. Therefore, it has certain necessity for the recognition of motor imagery to decompose the EEG signals into particular frequency bands. The common decomposition method include wavelet transform^[3], wavelet packet^[4], etc. Although these methods could achieve good results, there is the energy leakage phenomenon at the transition band of adjacent scales while the signal is decomposed using discrete wavelet^[5]. At the same time, the wavelet transform

will produce large aliasing phenomenon in the process of dual extraction^[6]. These defects will cause a certain degree of illusion in the feature extraction, and impact the performance of classifier. The Dual-tree complex wavelet transform (DTCWT), originally proposed by Kingsbury, is an enhancement of the conventional discrete wavelet transform (DWT)^[7]. DTCWT can circumvent the problem of the aliasing and energy leakage, and has some advantages, for example shift-invariance, multidimensional direction selectivity, and perfect reconstruction^[8]. The performance of feature extraction of DTCWT is superior to that of wavelet transform^[9-10].

In addition, the EEG signal is a typical nonlinear signal. The methods of feature extraction commonly transform the non-stationary and nonlinear EEG signal into stationary signal, but it has certain limitation. With the development of nonlinear theory, many nonlinear methods have been widely used in the feature extraction of EEG signal. Sample entropy (SampEn), proposed by Richman and Moorman^[11], is a method frequently used in nonlinear dynamics analysis. Compared with other nonlinear methods, for example Lyapunov index, information entropy, correlation dimension, and K entropy, SampEn can calculate steady estimation value with less data. It also has good anti-noise and anti-jamming ability. Song^[12] put forward an automatic epilepsy detection system which uses sample entropy as the only input feature, together with extreme learning machine (ELM) classification model. Results show that the proposed epilepsy detection approach achieves not only high detection accuracy but also a very fast computation speed. Han Ling^[13] presented a new approach to detect epileptic seizure. The new scheme was based on discrete wavelet transform (DWT) and sample entropy analysis of EEG signals. The result shows that the detection accuracy of the DWT sample entropy method is higher than the sample entropy methods.

In this work, a feature extraction method combining DTCWT and sample entropy is proposed for motor imagery EEG based BCI. DTCWT is used for decomposition and

*This work is supported by Zhejiang Provincial Natural Science Foundation of China under Grant LY14F030023 and National Natural Science Foundation of China under Grant 61372023 and 61201300.

reconstruction of EEG signals at the specific bands. Then the nonlinear features are extracted using sample entropy for the EEG signals recognition. This paper has been organized as follows. Section 2 introduces the theory of DTCWT and sample entropy, and describes the algorithm and experiments of feature extraction based on them. In Section 3, the evaluation procedure of recognition with SVM classifier and obtained experimental results are presented. Finally, some conclusions are included in Section 4.

2 Feature extraction based on DTCWT and SampEn

2.1 DTCWT

The DTCWT comprises of two real filters trees, *a* and *b*, as in Fig. 1, which produce the real and imaginary parts of complex coefficients. Odd and even length biorthogonal linear-phase filters are designed as the filter for the real and imaginary trees.

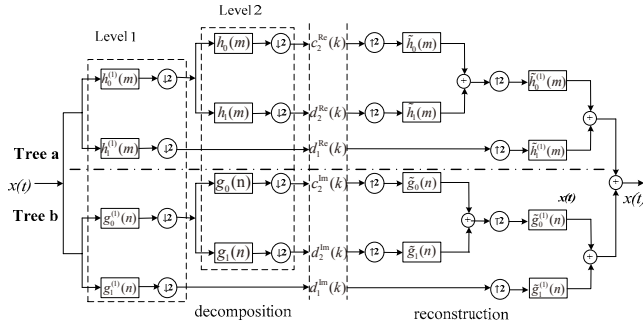


Fig. 1: The decomposition and reconstruction procedures of DTCWT

Below level 1, it can be found that the *tree b* samples do not interpolate midway between the *tree a* samples, because of the lower sampling rates at these levels. Therefore, shift invariance would not be achieved below level 1. To realize it at every level, the total delay difference for a given level and all previous levels must sum to one sample period at the input sample rate of the given level^[14]. Hence the filters below level 1 in one tree must provide delays that are half a sample different (at each filter's input rate) from those in the opposite tree. For linear phase filters, this requires odd-length filters in one tree and even-length filters in the other.

All the sampling rates in *Tree a* of Fig.1 can be doubled by eliminating the down-sampling by 2 after the level 1 filters, $h_0^{(1)}(m)$ and $h_1^{(1)}(m)$. This is equivalent to two parallel fully-decimated trees. The wavelet coefficient $d_j^{\text{Re}}(n)$ and scale coefficient $c_j^{\text{Re}}(n)$ in *Tree a* can be obtained as:

$$d_j^{\text{Re}}(n) = 2^{j/2} \int_{-\infty}^{+\infty} x(t) \psi_h(2^j t - n) dt \quad (1)$$

$$c_j^{\text{Re}}(n) = 2^{j/2} \int_{-\infty}^{+\infty} x(t) \phi_h(2^j t - n) dt \quad (2)$$

Likewise, the wavelet coefficient $d_j^{\text{Im}}(n)$ and scale coefficient $c_j^{\text{Im}}(n)$ in *Tree b* are:

$$d_j^{\text{Im}}(n) = 2^{j/2} \int_{-\infty}^{+\infty} x(t) \psi_g(2^j t - n) dt \quad (3)$$

$$c_j^{\text{Im}}(n) = 2^{j/2} \int_{-\infty}^{+\infty} x(t) \phi_g(2^j t - n) dt \quad (4)$$

In (1), (2), (3) and (4), n is sample number, j is scale factor, J is the maximum scale and $j=1,2,\dots,J$, $x(t)$ is sample signal. ψ_h, ϕ_h, ψ_g and ϕ_g represent the functions of wavelet transforms, satisfying the following relations:

$$\phi_h(t) = \sqrt{2} \sum_n h_0(n) \phi(2t - n) \quad (5)$$

$$\psi_h(t) = \sqrt{2} \sum_n h_1(n) \phi(2t - n) \quad (6)$$

$$\phi_g(t) = \sqrt{2} \sum_n g_0(n) \phi(2t - n) \quad (7)$$

$$\psi_g(t) = \sqrt{2} \sum_n g_1(n) \phi(2t - n) \quad (8)$$

where $h_0(n)$ and $h_1(n)$ represent low pass filter and high pass filter in real part, $g_0(n)$ and $g_1(n)$ represent low pass filter and high pass filter in imaginary part.

Integrating the correlation coefficient of real and imaginary parts, the wavelet coefficient $d_j^{(C)}(n)$ and scaling coefficient $c_j^{(C)}(n)$ in the decomposition procedure of DTCWT could be obtained as:

$$d_j^{(C)}(n) = d_j^{\text{Re}}(n) + i d_j^{\text{Im}}(n) \quad (9)$$

$$c_j^{(C)}(n) = c_j^{\text{Re}}(n) + i c_j^{\text{Im}}(n) \quad (10)$$

In the reconstruction procedure, the wavelet coefficient $d_j(t)$ and scaling coefficient $c_j(t)$ could be inverted as:

$$d_j(t) = 2^{(j-1)/2} \left[\sum_{n=-\infty}^{\infty} d_j^{\text{Re}}(n) \psi_h(2^j t - n) + \sum_{k=-\infty}^{\infty} d_j^{\text{Im}}(n) \psi_g(2^j t - k) \right] \quad (11)$$

$$c_j(t) = 2^{(j-1)/2} \left[\sum_{n=-\infty}^{\infty} c_j^{\text{Re}}(n) \phi_h(2^j t - n) + \sum_{k=-\infty}^{\infty} c_j^{\text{Im}}(n) \phi_g(2^j t - k) \right] \quad (12)$$

Finally, the signal $x(t)$ could be acquired using the outputs in (11) and (12):

$$x(t) = d_j(t) + c_j(t) \quad (13)$$

2.2 Sample Entropy

Entropy, as it relates to dynamical systems, reflects the rate of information production. Approximate entropy (ApEn) introduced by Pincus, is a set of measures of system complexity, which is easily applied to clinical cardiovascular and other time series^[15]. However, ApEn statistics often lead to inconsistent results. To overcome this defect, Richman and Moorman^[11] proposed a new and related complexity measure, sample entropy. The sample entropy value could be calculated as the following steps^[16]:

(1) Given a N points of time series $\{x_i\} : x(1), x(2) \dots x(N)$, take a set of m dimension vector $X(i)$ defined as:

$$X(i) = [x(i), x(i+1), \dots, x(i+m-1)] \quad (14)$$

(2) The distance between $X(i)$ and $X(j)$, $dis[X(i), X(j)]$, is defined as the largest absolute difference between two corresponding elements:

$$dis[X(i), X(j)] = \max_{k=0, \dots, m-1} |x(i+k) - x(j+k)| \quad (15)$$

(3) For the given threshold r ($r > 0$), count the number that $dis[X(i), X(j)]$ is less than r , written as $A_i^m(r)$, and then calculate the ratio of the number and distance number $N - m$ as:

$$B_i^m(r) = \frac{A_i^m(r)}{N-m} \quad (16)$$

(4) Calculate the average of all $B_i^m(r)$:

$$B^m(r) = \frac{1}{N-m+1} \sum_{i=1}^{N-m+1} B_i^m(r) \quad (17)$$

(5) Increment the dimension to $m+1$, and repeated step (1) ~ (4) to get $B^{m+1}(r)$. Thus, $B^m(r)$ represents the probability that two sequences will match for m points, whereas $B^{m+1}(r)$ represents the probability that two sequences will match for $m+1$ points.

(6) The sample entropy is defined by:

$$\text{SampEn}(N, m, r) = -\ln[B^{m+1}(r) / B^m(r)] \quad (18)$$

Sample entropy value is related to the value of sample entropy embedding dimension m and similar tolerance r . The value increases with the increase of m . The increase of r can reduces the noise influence, but will lose the details of time series. In general, $m=1$ or 2 , $r=0.1 \sim 0.25\sigma$, where σ is standard deviation of original data sequence $\{x_i\}$.

2.3 Data description

This paper adopts Datasets1 of BCI Competition IV^[17] provided by Berlin research group BCI in 2008 as an experiment data for the feature extraction and recognition. This dataset of EEG is collected from seven healthy subjects, wherein, five subjects participating in motor imagery tasks of left and right hand, two subjects choosing sessions of left hand and foot.

In the first two runs, arrows pointing left, right, or down were presented as visual cues on a computer screen. Cues were displayed for a period of 4 s during which the subject was instructed to perform the cued motor imagery task from left hand, right hand or foot. Specifically, subject “a” chose { left, foot}, “b” chose { left, right}, “f” chose { left, foot}, “g” chose { left, right}. These periods were interleaved with 2 s of blank screen and 2 s with a fixation cross shown in the center of the screen. The fixation cross was superimposed on the cues, i.e., it was shown for 6 s (see Fig. 2). Each subject performed a total of 200 motor imagery tasks. Signals from 59 EEG positions were measured that were most densely distributed over sensorimotor areas. Signals were band-pass filtered between 0.05 and 200 Hz. Also a version of the data was provided that was sub sampled at 100 Hz. In the paper the calibration data of subjects “a” “b” “f” “g” are selected for classification experiment.

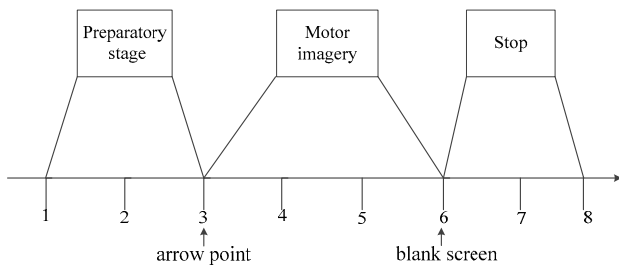


Fig. 2: Sequence diagram of motor imagery task

2.4 Feature extraction method combined DTCWT and Sample Entropy

Feature extraction is a critical step in the pattern recognition. It not only affects the design of classifier, but also is related to the effectiveness of classification. A new feature extraction method combining DTCWT and SampEn is proposed in this paper. Firstly, the motor imagery EEG signals are decomposed in different frequency bands by DTCWT. Then, the useful rhythm waves corresponding to the ERD / ERS phenomenon are chosen and reconstructed. Finally, the nonlinear features are extracted from EEG reconstructed using sample entropy. In other hand, C3 and C4 electrodes are located on the primary sensorimotor area of cortex, which can best reflect the brain activity during the motor imagery. Therefore, to reduce the number of features, only channels C3 and C4 among 59 channels are employed for feature extraction. DTCWT could be used for decomposition of EEG signal of C3, C4 channel, and the number of decomposition will depend on the useful components and sampling frequency of specific signal.

Assuming the sampling frequency of EEG signal acquired is f_s , and the level of decomposition is L , the complex coefficient of low-frequency subband cA_L and the complex coefficient of high-frequency subband $cD_L, cD_{L-1}, \dots, cD_1$ can be obtained by DTCWT decomposition. Then the signal variables $A_L, D_L, D_{L-1}, \dots, D_1$ are reconstructed with the frequency band $[0, f_s / 2^{L+1}]$, $[f_s / 2^{L+1}, f_s / 2^L]$, $[f_s / 2^L, f_s / 2^{L-1}]$, $\dots$, $[f_s / 2^2, f_s / 2]$. Then the signal variables corresponding to the frequency band of alpha rhythm and beta rhythm are chosen to reconstruct the signals for the feature extraction.

The sampling frequency of the signal in Datasets 1 is 100 Hz, and the level of decomposition is 3. Thus, the corresponding sub frequency band of each component is shown in Table 1.

Table 1: The corresponding sub frequency band of each component of DTCWT

Component	A3	D3	D2	D1
f (Hz)	0~6.25	6.25~12.5	12.5~25	25~50

Table 1 shows that D2 and D3 are near in the frequency band range of the alpha rhythm (8 ~ 15 Hz) and beta rhythm (15 ~ 30 Hz). Therefore, the components D2 and D3 are selected for signal reconstruction. Then feature is extracted from the signal on these frequency bands respectively using sample entropy. In the calculation of the sample entropy, parameter m and r values in this simulation are set as 2 and 0.25σ . Therefore, four features can be obtained for a motor imagery task from these two channels (C3 and C4). The SampEn feature distributions of two imagery task of each subject are given according to the alpha and beta rhythm, as shown in Fig. 3.

In order to compare with the method that directly extracts the feature from raw EEG signals, which are not decomposed and reconstructed using DTCWT, SampEn features of these signals are also extracted as the two dimensional feature vectors. As shown in Fig. 4, it is

obvious that the distribution of features obtained from raw signals overlap more than from reconstructed signals between two classes. Moreover, the feature extracted using proposed method has more obvious separability and is easier for further classification.

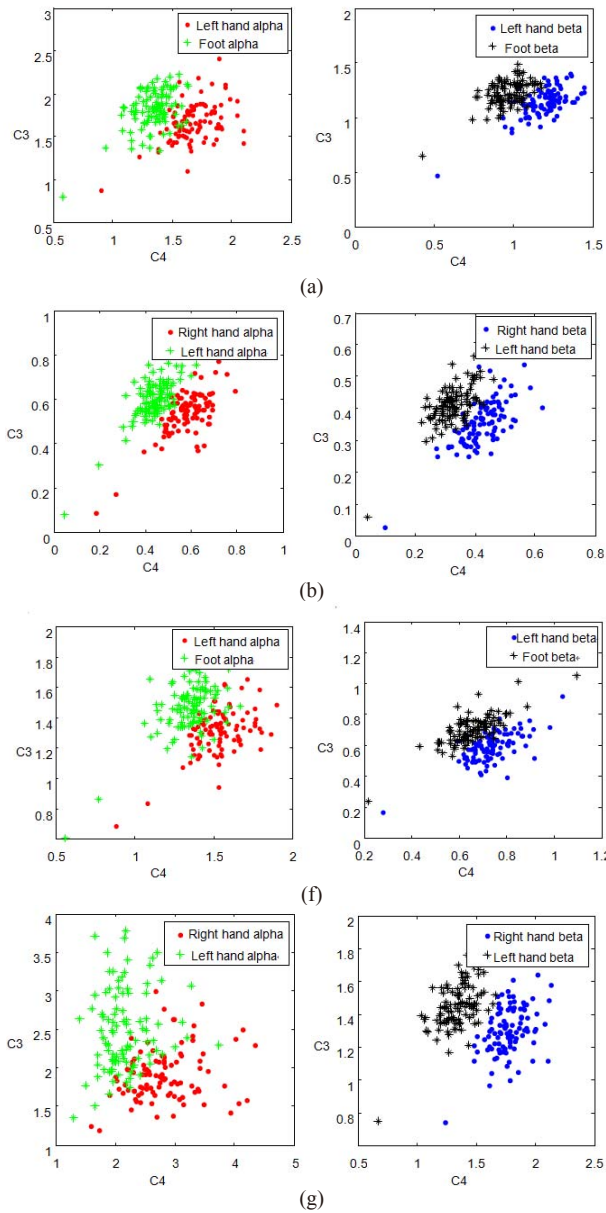


Fig. 3: The sample entropy features of alpha and beta rhythm of subjects a, b, f and g

3 Experiments of Classification

In order to assess the performance of the feature extraction method for classifying motor imagery tasks, this paper adopts the support vector machine (SVM) as the classifier to carry out the classification experiments on the Datasets1 of BCI Competition IV. SVM has been successfully used in many classification applications. It is a kind of pattern recognition method based on statistical learning theory, and can effectively avoid the defects of traditional classification methods, such as over learning, dimension disaster and local minima. Besides, it still has good generalization ability under the condition of small sample^[18].

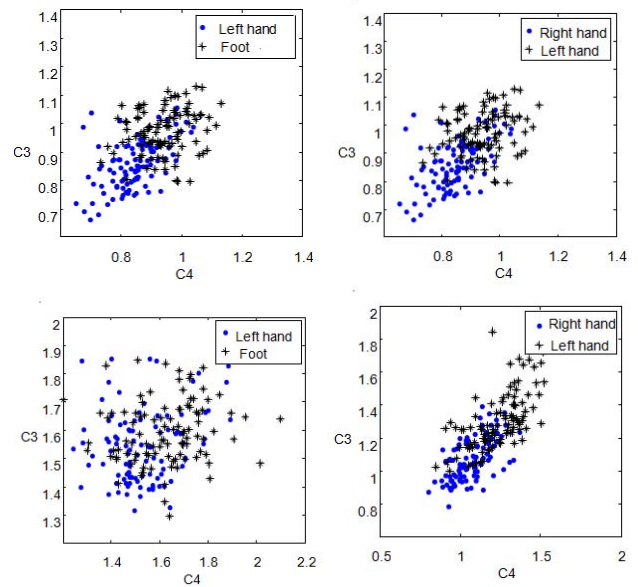


Fig. 4: The sample entropy features of raw EEG signals of subjects a, b, f and g

For each motor imagery task of each subject, 50 EEG signals are randomly selected as the training samples, and the other 50 signals selected as the test samples. C3 and C4 channel signals are selected from each sample for classification. A four dimensional feature vector can be extracted using the method of section 2.4. The SVM classifier with Gauss kernel function is trained for each subject using fivefold cross-validation method to select the optimal penalty parameter and kernel parameter. Finally, SVM classifier is adopted to classify the test sample set.

The multiple classification experiment results of the four subjects in the Datasets1 are shown in Fig. 5, where the vertical axis represents the classification rate, and abscissa represents the four subjects a, b, f and g. It can be seen that the median of the recognition rate of four subjects are more than 0.85, which means that the results of more than half of the classification experiments exceed 85%.

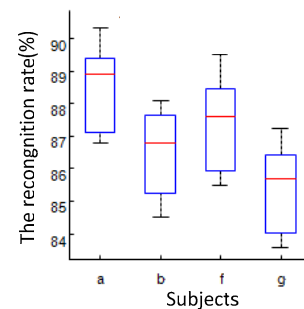


Fig. 5: The classification results of each subject

In order to further verify the validity of the proposed method, this method is compared with other commonly used method, which have achieved remarkable results on the same datasets, such as the common spatial pattern (CSP) method, and its improved method filter group common spatial pattern (FBCSP), which combines a filter bank framework with CSP, and optimum spatio-spectral filtering network (OSSFN)^[19]. The recognition rates of these

methods and proposed method on the same subjects are given in Table 2.

Table 2: The recognition rates with different methods

Subject	Feature extraction method			
	CSP	FBCSP	OSSFN	DTCTWT SampEn
a	67.1	66.5	89.8	88.9
b	76.6	87.3	86.8	86.8
f	65.3	87.2	93.0	87.6
g	74.6	82.5	92.4	85.7
Average	70.9	80.88	90.5	87.3

Compared with the basic CSP method and FBCSP method, DTCTWT SampEn method acquire significantly higher classification accuracy in most cases. Compared to the OSSFN method, there is still a gap on the classification accuracy, especially in subjects f and g. Although OSSFN method can obtain higher classification accuracy, but it uses all 59 channels of EEG signal, which means the high requirements of the data acquisition and computation. The proposed method uses only two channels of all 59 channels, therefore the data acquisition condition and computational requirements are reduced obviously, which is helpful to improve the practicability of BCI system.

4 Conclusion

This paper presents a method of feature extraction combined with DTCTWT and SampEn. This method decomposes the motor imagery EEG signals using DTCWT, and then abstracts and reconstructs the rhythm signals. The features are extracted using sample entropy. Because the proposed method considers the nonlinear characteristics and ERD/ERS phenomenon of motor imagery EEG signals, it could obtain high classification accuracy in motor imagine classification experiment with SVM classifier. It also shows that the combination with DTCTWT and SampEn can effectively characterize complexity feature of motor imagery EEG signals. Therefore, the method provides a new study idea for the EEG classification.

References

- [1] J. Wolpaw, N. Birbaumer, J. McFarland, et al. Brain-computer interfaces for communication and control, *Clinical Neurophysiol*, 113(6):767-791, 2002.
- [2] D. Valbuena, M. Cyriacks, O. Friman, et al. Brain-computer interface for high-level control of rehabilitation robotic systems, in *Proceedings of IEEE 10th International Conference on Rehabilitation Robotics*, 2007: 619-625.

- [3] H. Adeli, Z. Zhou, and N. Dadmehr, Analysis of EEG records in an epileptic patient using wavelet transform, *Journal of neuroscience methods*, 123(1): 69-87, 2003.
- [4] G. Yen and K. Lin C, Wavelet packet feature extraction for vibration monitoring, *IEEE Transactions on Industrial Electronics*, 47(3): 650-667, 2000.
- [5] Z. Peng, R. Jackson, F. Chu, et al. On the energy leakage of discrete wavelet transform, *Mechanical Systems and Signal Processing*, 23(2): 330-343, 2009.
- [6] I. Selesnick, R. Baraniuk, and N. Kingsbury, The dual-tree complex wavelet transform, *IEEE Signal Processing Magazine*, 22(6): 123-151, 2005.
- [7] N. Kingsbury, The dual-tree complex wavelet transform: a new technique for shift invariance and directional filters, *Philosophical Transactions of the Royal Society of London, Series A: Mathematical, Physical and Engineering Sciences*, 8: 2543-2560, 1998.
- [8] K. Chaudhury and M. Unser, On the shiftability of dual-tree complex wavelet transforms, *IEEE Transactions on Signal Processing*, 58(1): 221-232, 2010.
- [9] M. Thomas, M. Das, and S. Ari, Classification of cardiac arrhythmias based on dual tree complex wavelet transform, in *Proceedings of IEEE 2014 International Conference on Communications and Signal Processing*, 2014: 729-733.
- [10] M. Shahid and S. Gupta, Novel masks for multimodality image fusion using DTCWT, in *Proceedings of 9th International Conference on Information Fusion*, 2006:1-5.
- [11] J. Richman and J. Moorman, Physiological time-series analysis using approximate entropy and sample entropy, *American Journal of Physiology-Heart and Circulatory Physiology*, 278(6): 2039-2049, 2000.
- [12] Y. Song, J. Crowcroft, and J. Zhang, Automatic epileptic seizure detection in EEGs based on optimized sample entropy and extreme learning machine, *Journal of neuroscience methods*, 210(2): 132-146, 2012.
- [13] L. Han, H. Wang, C. Liu, et al. Epileptic seizure detection using wavelet transform based sample entropy and support vector machine, in *Proceedings of IEEE 2012 International Conference on Information and Automation*, 2012: 759-762.
- [14] N. Kingsbury, Complex wavelets for shift invariant analysis and filtering of signals, *Applied and computational harmonic analysis*, 10(3): 234-253, 2001.
- [15] S. Pincus, Approximate entropy (ApEn) as a complexity measure, *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 5(1): 110-117, 1995.
- [16] J. Lai, Research on natural language recognition algorithm based on sample entropy, *Applied Mechanics and Materials*, 539: 185-189, 2014.
- [17] M. Tangermann, K. Müller R, A. Aertsen, et al. Review of the BCI competition IV, *Frontiers in neuroscience*, 6(55):1-31, 2012.
- [18] J. Shi, W. Lee, Y. Liu, et al. Forecasting power output of photovoltaic systems based on weather classification and support vector machines, *IEEE Transactions on Industry Applications*, 48(3): 1064-1069, 2012.
- [19] H. Zhang, Z. Chin, K. Ang, et al. Optimum spatio-spectral filtering network for brain-computer interface, *IEEE Transactions on Neural Networks*, 22(1): 52-63, 2011.